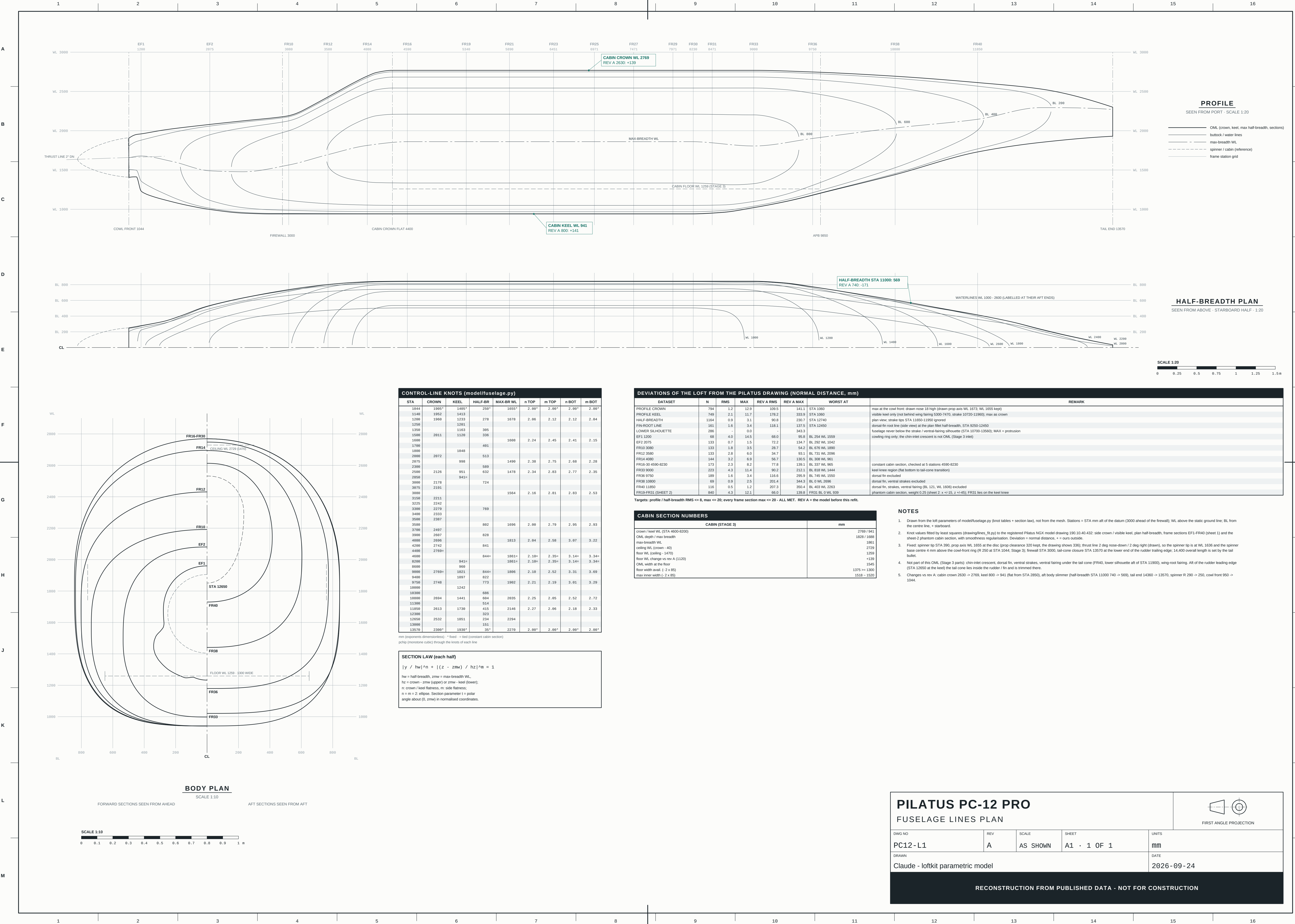




CONTROL-LINE KNOTS (model/fuselage.py)									
STA	CROWN	KEEL	HALF-BR	MAX-BR WL	n TOP	m TOP	n BOT	m BOT	
1844	1865*	1405*	258*	1655*	2.06*	2.08*	2.08*	2.08*	
1148	1852	1413							
1209	1906	1233	278	1678	2.06	2.12	2.12	2.04	
1259		1261							
1358		1163	305						
1508	2811	1126	336						
1608				1608	2.24	2.45	2.41	2.15	
1768			461						
1808		1848							
2088									
2975	2872	998	513	1400	2.38	2.75	2.68	2.28	
2368			589						
2568	2126	951	632	1478	2.34	2.83	2.77	2.35	
2658		941=							
3888	2178		724						
3875	2191								
3888				1564	2.16	2.81	2.83	2.53	
3158	2211								
3225	2242								
3308	2279		769						
3408	2333								
3508	2387								
3588			882	1696	2.08	2.79	2.95	2.93	
3708	2487								
3908	2687		828						
4008	2696			1813	2.04	2.58	3.07	3.22	
4208	2742		841						
4408	2769=								
4608		941=	844=	1861=	2.18=	2.35=	3.14=	3.34=	
8208		960		1861=	2.18=	2.35=	3.14=	3.34=	
9008	2769=	1021	844=	1806	2.18	2.52	3.31	3.69	
9408		1097	822						
9758	2748		773	1982	2.21	2.19	3.01	3.29	
10008		1242							
10308			686						
10808	2684	1441	604	2035	2.25	2.05	2.52	2.72	
11308			514						
11858	2613	1730	415	2146	2.27	2.06	2.18	2.33	
12308			323						
12658	2532	1851	234	2294					
13008			152						
13578	2388*	1938*	35*	2270	2.00*	2.00*	2.00*	2.00*	

mm (exponents dimensionless) * = fixed - = tied (constant cabin section)
pchip (monotone cubic) through the knots of each line

SECTION LAW (each half)

$$|y| / h^w |^n + |(z - zmw) / hz|^m = 1$$

hw = half-breadth, zmw = max-breadth WL,
hz = crown - zmw (upper) or zmw - keel (lower);
n: crown / keel flatness, m: side-flatness;
n = m = 2: ellipse. Section parameter t = polar
angle about (0, zmw) in normalised coordinates.

DEVIATIONS OF THE LOFT FROM THE PILATUS DRAWING (NORMAL DISTANCE, mm)							REMARK
DATASET	N	RMS	MAX	REV A RMS	REV A MAX	WORST AT	
PROFILE CROWN	794	1.2	12.9	109.5	141.1	STA 1060	max at the cowl front: drawn nose 18 high (drawn prop axis WL 1673; WL 1655 kept)
PROFILE KEEL	749	2.1	11.7	178.2	333.9	STA 1060	visible keel only (not behind wing fairing 5300-7470; strake 10720-11960); max as crown
HALF-BREADTH	1164	0.9	3.1	90.8	230.7	STA 12740	plan view; strake tips STA 11650-11950 ignored
FIN-ROOT LINE	161	1.6	3.4	118.1	137.5	STA 12450	dorsal-fin root line (side view) at the plan fillet half-breadth; STA 9250-12450
LOWER SILHOUETTE	286	-	0.0	-	343.3		fuselage never below the strake / ventral-fairing silhouette (STA 10700-13560); MAX = protrusion
EF1 1200	68	4.0	14.5	68.0	95.8	BL 254 WL 1559	cowling ring only; the chin-inlet crescent is not OML (Stage 3 inlet)
EF2 2075	133	0.7	1.5	72.2	134.7	BL 292 WL 1042	
FR10 3080	132	1.8	3.5	28.7	54.2	BL 676 WL 1890	
FR12 3580	133	2.8	6.0	34.7	93.1	BL 731 WL 2096	
FR14 4080	144	3.2	6.9	56.7	130.5	BL 308 WL 961	
FR16-30 4590-8230	173	2.3	8.2	77.8	139.1	BL 337 WL 965	constant cabin section, checked at 5 stations 4590-8230
FR33 9080	225	4.3	11.4	90.2	212.3	BL 618 WL 1444	keel knee region (flat bottom to tail-cone transition)
FR38 9750	189	1.6	3.4	116.6	295.9	BL 745 WL 1550	dorsal fin excluded
FR38 10800	69	0.9	2.5	201.4	344.3	BL 0 WL 2696	dorsal fin, ventral strakes excluded
FR40 11850	116	0.5	1.2	207.3	350.4	BL 403 WL 2263	dorsal fin, strakes, ventral fairing (BL 121; WL 1606) excluded
FR19-FR31 (SHEET 2)	840	4.3	12.1	66.0	139.8	FR31; BL 0 WL 939	phantom cabin section, weight 0.25 (sheet 2 - x +/-15, z +/-45); FR31 lies on the keel knee

Targets: profile / half-breadth RMS <= 8, max <= 20; every frame section max <= 20 - ALL MET. REV A = the model before this refit.

CABIN SECTION NUMBERS	
CABIN (STAGE 3)	mm
crown / keel WL (STA 4600-8200)	2769 / 941
OML depth / max breadth	1828 / 1688
max-breadth WL	1861
ceiling WL (crown - 40)	2729
floor WL (ceiling - 1470)	1259
floor WL change vs rev A (1120)	+139
OML width at the floor	1545
floor width avail. (- 2 x 85)	1175 >= 1300
max inner width (- 2 x 85)	1518 - 1520

NOTES

- Drawn from the loft parameters of model/fuselage.py (knot tables + section law), not from the mesh. Stations = STA mm aft of the datum (3000 ahead of the firewall); WL above the static ground line; BL from the centre line, + starboard.
- Knot values fitted by least squares (drawing/lines, fit.py) to the registered Pilatus NGX model drawing 190.10.40.432: side crown / visible keel, plan half-breadth, frame sections EF1-FR40 (sheet 1) and the sheet-2 phantom cabin section, with smoothness regularisation. Deviation + normal distance, + = ours outside.
- Fixed: spinner tip STA 300, prop axis WL 1655 at the disc (prop clearance 320 kept, the drawing shows 336); thrust line 2 deg nose-down / 2 deg right (drawn), so the spinner tip is at WL 1636 and the spinner base centre 4 mm above the cowl-front ring (R 250 at STA 1044; Stage 3); firewall STA 3000, tail-cone closure STA 13570 at the lower end of the rudder trailing edge; 14,400 overall length is set by the tail bullet.
- Not part of this OML (Stage 3 parts): chin-inlet crescent, dorsal fin, ventral strakes, ventral fairing under the tail cone (FR40, lower silhouette aft of STA 11900), wing-root fairing. Aft of the rudder leading edge (STA 12650 at the keel) the tail cone lies inside the rudder / fin and is trimmed there.
- Changes vs rev A: cabin crown 2630 -> 2769, keel 800 -> 941 (flat from STA 2850), aft body slimmer (half-breadth STA 11000 740 -> 569), tail end 14360 -> 13570, spinner R 290 -> 250, cowl front 950 -> 1044.

PILATUS PC-12 PRO

FUSELAGE LINES PLAN

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L1	A	AS SHOWN	A1 · 1 OF 1	mm
DRAWN				DATE
Claude - loftkit parametric model				2026-09-24

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION